

DBMS Project Synopsis

Blood Bank Management System

Course Name & Code: UCS310 – Database Management Systems

Degree & Year: B.Tech (2nd Year)

Department / Institute: Thapar Institute of Engineering and Technology

Group Members:

- Krish Kataria – Roll No: 1024030051
- Krish Sareen – Roll No: 1024030053
- Arushi Yadav – Roll No: 1024030052

Lab Instructor Name: _____

Academic Year: 2025–26

1. Introduction

The Blood Bank Management System is a database-driven application designed to maintain records of donors, patients, hospitals, blood inventory, donations, and blood requests. This system improves data accuracy, consistency, and retrieval speed. It replaces traditional file-based systems which often suffer from redundancy and slow access.

2. Problem Statement

Manual systems for managing blood banks are slow, error-prone, and difficult to maintain. There is difficulty in tracking available blood units, managing donor records, and maintaining request history. This project aims to create a structured database system to manage donors, patients, blood inventory, and hospital requests efficiently.

3. Objectives of the Project

- To design a database using ER data modeling
- To convert ER model into relational tables
- To apply normalization up to 3NF
- To implement database using SQL
- To use PL/SQL features like procedures, triggers, and functions
- To maintain data integrity using constraints

4. Scope of the Project

Users: Admin, Hospital Staff, Blood Bank Staff

Modules:

- Donor Management
- Patient Management
- Hospital Management
- Blood Inventory Tracking
- Blood Request Processing
- Donation Management
- Reward Management

5. Proposed System Description

The system stores records of donors, patients, hospitals, blood donations, inventory, and rewards. It enables real-time tracking of blood availability, updates inventory after each donation, and reduces stock when requests are fulfilled.

6. Database Design

Entities:

Patient(patient_id, name, blood_type, diagnosis, contact, hospital_id)

Hospital(hospital_id, name, address, contact)

Donor(donor_id, name, blood_type, contact, address, last_donation, reward_points)
Blood_Inventory(inventory_id, blood_type, quantity, expiry_date, status)
Blood_Request(request_id, patient_id, inventory_id, quantity, request_date, status)
Blood_Donation(donation_id, donor_id, inventory_id, donation_date, blood_type, quantity, status)
Donor_Rewards(reward_id, donor_id, points, reward_type, earned_date)

Relationships & Cardinality:

- One hospital admits many patients (1:M)
- One patient can make multiple requests (1:M)
- One donor can donate multiple times (1:M)
- Inventory fulfills blood requests

7. Normalization

All tables are normalized up to 3NF to remove redundancy and ensure data integrity.

8. Database Implementation

SQL:

- CREATE, INSERT, UPDATE, DELETE operations
- Joins, Subqueries, Aggregate Functions
- Views and Constraints

PL/SQL:

- Stored Procedures
- Functions
- Triggers
- Cursors
- Exception Handling

9. Transaction Management & Concurrency

- COMMIT after successful operations
- ROLLBACK in case of errors
- SAVEPOINT for safe updates

Ensures ACID properties and data consistency.

10. Tools & Technologies Used

- Oracle / MySQL
- SQL / PL-SQL
- SQL Developer / MySQL Workbench

11. Expected Outcomes

- Fully normalized database
- Efficient blood stock tracking
- Fast query execution
- Automated updates using triggers
- Improved data integrity
- Real-time tracking of blood availability